IBM Applied Data Science Capstone

SpaceX Falcon 9 First-Stage Landing Prediction

Revised submission report • based on the uploaded cleaned dataset and freshly computed analysis

Implementation status: The analysis, SQL outputs, model evaluation, confusion matrices, charts, and Folium HTML were generated from the uploaded CSV. API collection and web scraping are included as runnable notebook cells/scripts but require a live internet run; the Dash app source is provided and must be launched locally for authentic screenshots. These outstanding steps are explicitly identified rather than presented as completed.

1. Executive summary

The cleaned dataset contains 90 launch records and 18 columns. There are 60 successful outcomes (66.7%) and 30 unsuccessful outcomes (33.3%). Two baseline classifiers were evaluated using a stratified 75/25 train/test split with random_state=42. The Decision Tree achieved accuracy 87.0% and F1 0.903; Logistic Regression achieved recall 100% on this test split. The project combines EDA, SQL aggregation, an interactive map, and predictive classification.

2. Introduction and analytical objective

Falcon 9 booster recovery supports reuse and launch-cost reduction. This educational analysis asks whether historical launch attributes are associated with first-stage landing outcomes. The target variable `Class` is coded 1 for successful landing and 0 for unsuccessful landing. The model is retrospective and should not be used as a flight-safety or operational decision system.

3. Data collection and wrangling methodology

The supplied cleaned CSV is the source used for all numerical results in this report. The data was loaded into pandas, columns were inspected, missing values and duplicate rows were counted, and the target distribution was checked. The dataset has 90 rows, 18 columns, zero exact duplicate rows, and 26 missing values in `LandingPad`; other columns have no missing values in this file. Missing values are handled inside model pipelines rather than by fitting preprocessing on the full dataset. The final notebook now includes dedicated API collection and web-table scraping cells; run them with internet access, save their outputs, and record retrieval date and row counts before claiming those collection steps are complete.

4. EDA and data visualization methodology

The EDA evaluates outcome balance, launch-site success rates, payload mass distributions by outcome, and observed success rates across flight-number groups. Bar charts show counts and site comparisons; histograms show payload overlap; a line chart shows the flight-group trend. Group sizes and sample limitations must accompany comparisons, and descriptive associations should not be interpreted as causal effects.

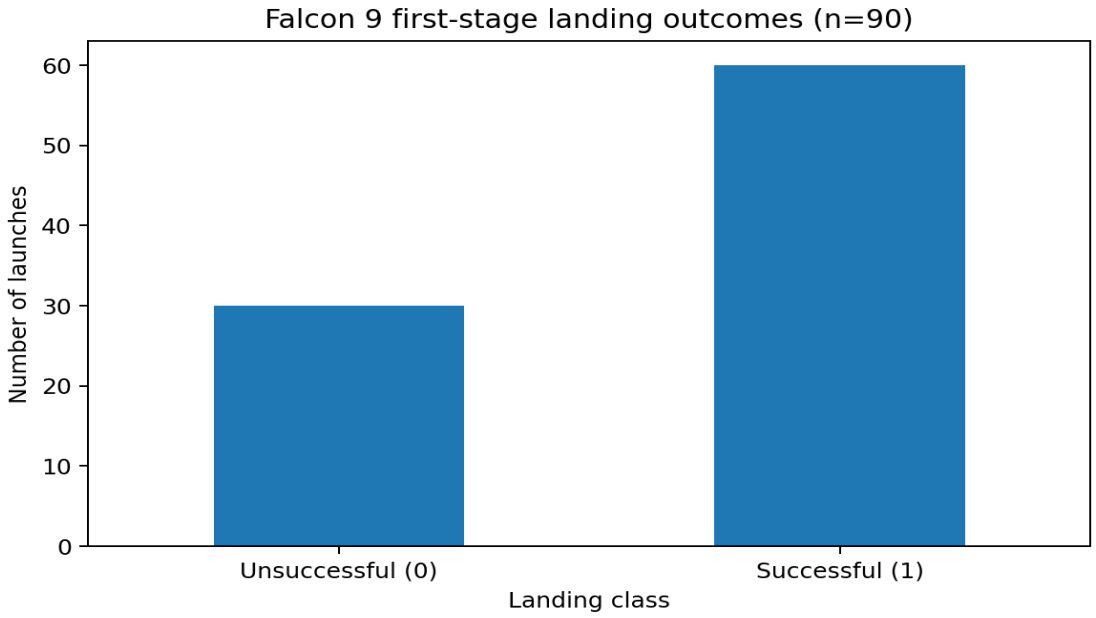


Figure 1. Landing outcome distribution: 60 successful and 30 unsuccessful outcomes.

4. EDA results (continued)

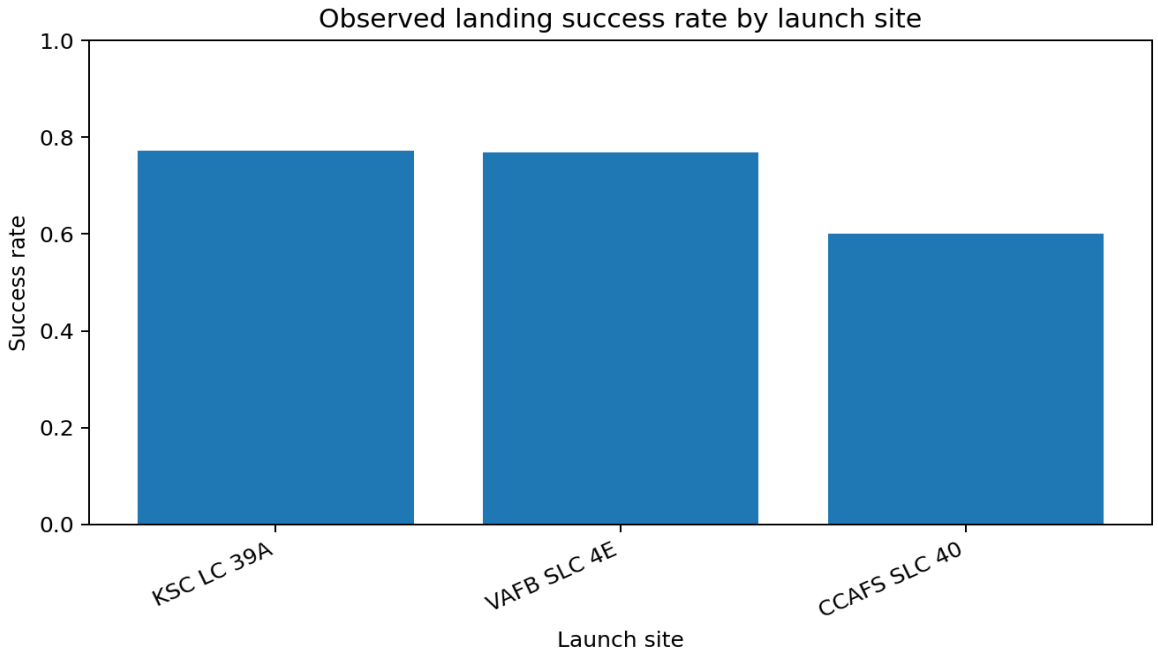


Figure 2. Observed success rate by launch site. Site-level differences should be interpreted with sample sizes and mission mix in mind.

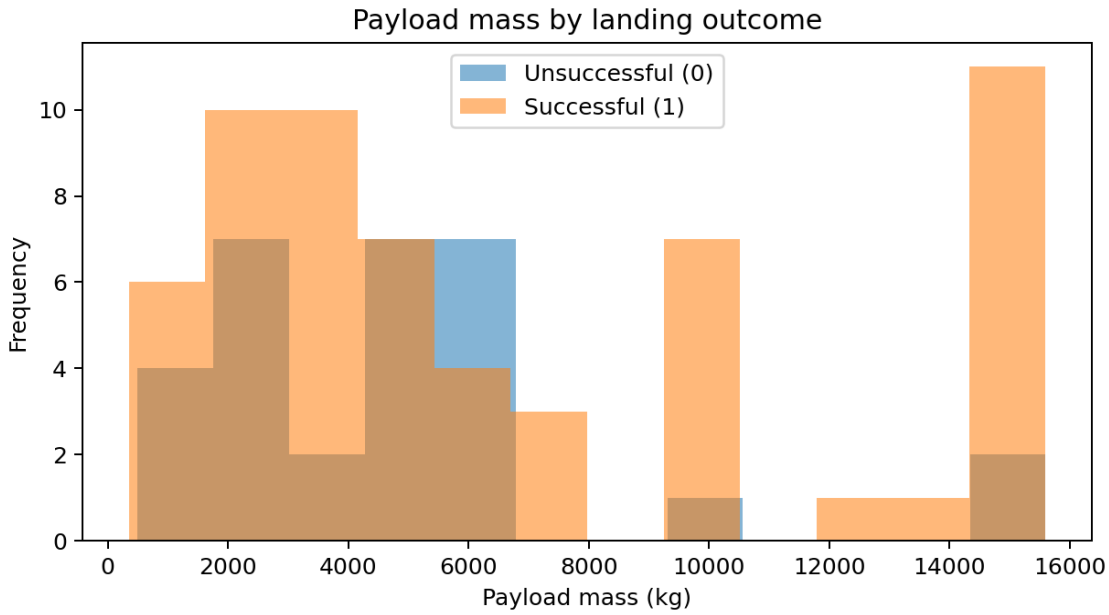


Figure 3. Payload mass distributions overlap across classes, so payload mass alone is not a deterministic predictor.

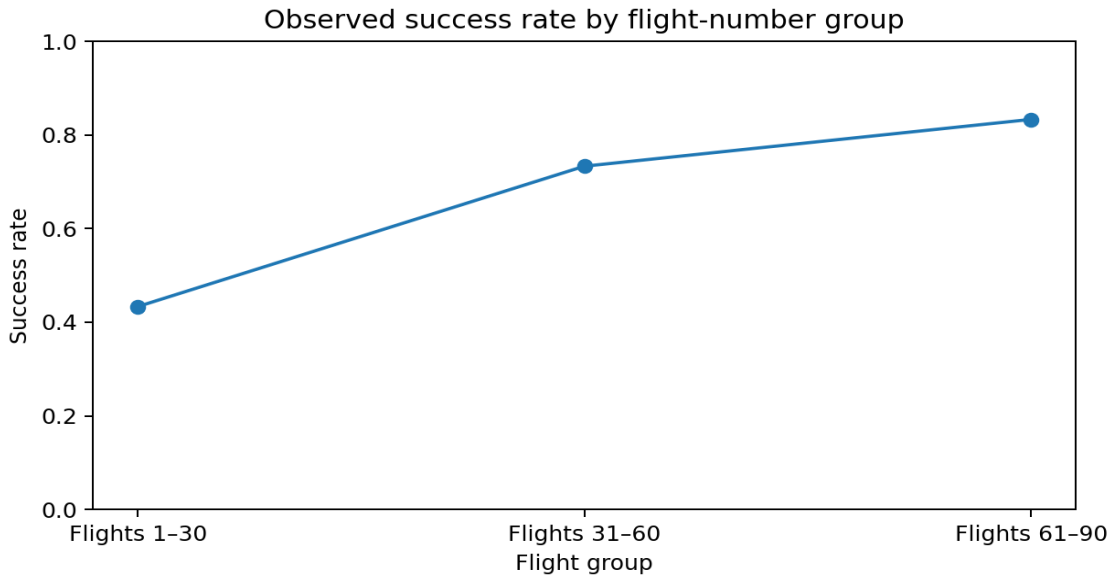


Figure 4. Observed success rates increase across the three flight-number groups in this sample; the trend does not prove causation.

5. EDA with SQL — queries and executed results

The uploaded CSV was loaded into SQLite and the following queries were executed: outcome counts, launches by site, success rate by site, and payload summary by outcome. The query text and CSV output files are included in the artifact package.

Outcome counts

Class	launches	pct
0	30	33.3
1	60	66.7

Launches by site

LaunchSite	total_launches
CCAFS SLC 40	55
KSC LC 39A	22
VAFB SLC 4E	13

Success rate by site

LaunchSite	total_launches	success_rate
KSC LC 39A	22	0.773
VAFB SLC 4E	13	0.769
CCAFS SLC 40	55	0.6

Payload summary by outcome

Class	records_with_payload	mean_payload_kg	min_payload_kg	max_payload_kg
0	30	4784.56	500.0	15600.0
1	60	6765.16	350.0	15600.0

Interpretation

The dataset contains 30 unsuccessful and 60 successful launches. CCAFS SLC 40 has the largest sample (55 launches) and an observed success rate of 0.600; KSC LC 39A has 22 launches and a rate of 0.773; VAFB SLC 4E has 13 launches and a rate of 0.769. Mean payload mass is 4,784.56 kg for Class 0 and 6,765.16 kg for Class 1 in this sample. These are descriptive statistics, not causal estimates.

6. Folium map

The generated `web/spacex_launch_sites.html` is an interactive Folium map with clustered site markers and pop-ups containing the site name, number of launches, and observed success rate. Open the HTML file in a browser to interact with the map. Include a genuine browser screenshot with a marker pop-up in the final repository evidence.

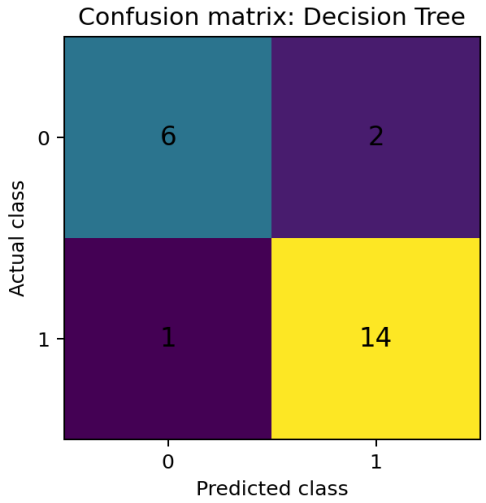
7. Predictive analysis (classification)

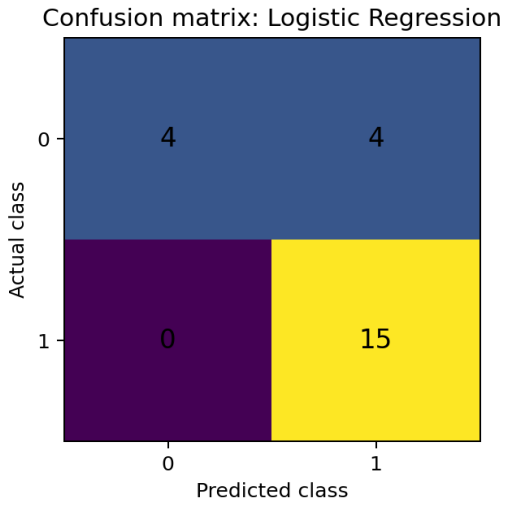
The notebook uses a stratified 75/25 train/test split with random_state=42 and compares Logistic Regression with a Decision Tree of maximum depth 5. Numeric fields are median-imputed and scaled; categorical fields are most-frequent-imputed and one-hot encoded within each training pipeline. Candidate features include FlightNumber, PayloadMass, Flights, GridFins, Reused, Legs, Block, ReusedCount, Orbit, LaunchSite, and Serial. Before using this as a prospective prediction model, review whether each field is available at prediction time and whether any variables encode post-outcome information.

Model	Accuracy	Precision	Recall	F1
Decision Tree	0.870	0.875	0.933	0.903
Logistic Regression	0.826	0.789	1.000	0.882

8. Predictive results and confusion matrices

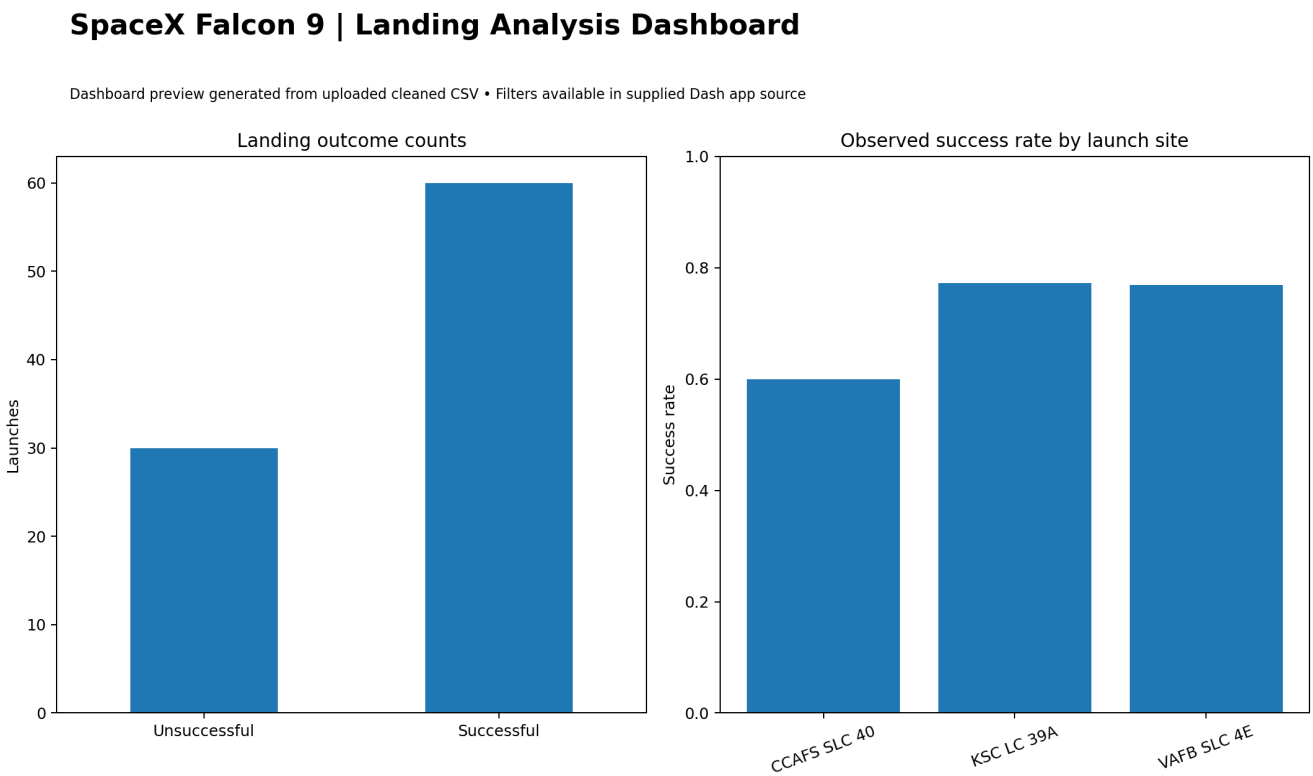
The confusion matrices below were calculated from actual held-out predictions in the fresh run. Rows represent actual classes [0, 1]; columns represent predicted classes [0, 1]. Decision Tree: [[6, 2], [1, 14]]. Logistic Regression: [[4, 4], [0, 15]]. The Decision Tree performs better in accuracy and F1 in this split; Logistic Regression detects all positive-class examples in the test set but generates more false positives.





9. Plotly Dash dashboard

The included `web/spacex_dash_app.py` defines launch-site and orbit dropdown filters and three interactive views: outcome counts, payload mass by outcome, and observed success rate by site. Run the app with `python web/spacex_dash_app.py` after installing dependencies, then open `http://127.0.0.1:8050`. The figure below is a static preview generated from the uploaded dataset to show the dashboard's analytical content; it is not a screenshot of a running Dash server. The rubric specifically asks for a live app screenshot and a second screenshot showing a changed filter, so capture these after running the app locally.



10. API collection and web scraping

The notebook and scripts include a SpaceX API v4 query workflow that saves raw JSON and a parsed CSV, plus a `pandas.read_html` workflow that lists candidate historical launch tables and saves them for inspection. Their network calls have not been executed in this environment, so this report does not claim API or scraping results. Run both with internet access, save the terminal output and source artifacts, inspect table choice/columns, and add the evidence to GitHub.

11. Conclusion, creativity, and limitations

The sample shows 66.7% successful landing outcomes. Decision Tree is the stronger of the two recorded baselines by accuracy and F1 in the held-out split, while Logistic Regression has higher recall. Creative extensions include an interactive geographic map, a filterable dashboard, and an auditable SQL/model pipeline. The dataset is small and a random split may overstate performance on future launches; time-aware validation and a careful leakage review are recommended. API/scraping outputs and live Dash screenshots remain required evidence.

12. Final submission checklist

Included: updated final notebook, cleaned dataset, data-quality report, SQL statements and result CSVs, EDA charts, model metrics and exact confusion matrices, Folium HTML, Dash source, API/scraping scripts, README, and requirements. **Still required from your computer:** execute API collection and scraping, run Dash and capture authentic screenshots, upload all files to GitHub, and paste the repository URL into the submitted PDF if the rubric expects it. This report cannot guarantee a particular grade.